

BCS Practitioner Certificate in Requirements Engineering

Sample Paper

Record your surname / last / family name and initials on the answer sheet.

Sample paper only 40 multiple-choice or multiple response questions – 1 mark is awarded to each question. There are no trick questions.

A number of possible answers are given for each question, indicated by either **A, B, C or D**. Your answers should be clearly indicated on the answer sheet.

Pass mark is 26/40.

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This professional certification is not regulated by the following United Kingdom Regulators
- Ofqual, Qualifications in Wales, CCEA or SQA

- 1 A practice guide for requirements elicitation offers the following information:

Disadvantages:

- Likely to be particularly time consuming to conduct.
- Difficult to separate fact from opinion.

Advantages:

- Provides an opportunity to build rapport.
- Provides an opportunity to explore various perspectives of the business environment.

Which elicitation method is being referred to?

- A Interviews.
- B Workshops.
- C Focus Groups.
- D Observation.

- 2 Woody's Bespoke Furniture has engaged a Business Analyst to define its requirements for automating currently paper-based business processes. The documents that they wish to automate include the work requisition for craftsmen and the customer invoice.

Which of the following issues **SHOULD** the class model clarify?

- A Why a work requisition must be raised for a piece of furniture.
- B Who raises a customer invoice.
- C How many craftsmen can work on a piece of furniture.
- D When a customer invoice is sent to a customer.

- 3** Topps Taxi service is in the process of moving its booking operations to a new online system. A meeting is taking place to prioritise the requirements for inclusion in the first release of the new application, which will take place in January and the second release is planned for April.

John is the owner of Topps Taxi and makes his requirements clear to the development team but later explains that having the ability to integrate airport arrival information into his booking system would enable his drivers to receive alerts when flights are delayed or cancelled, saving wasted journeys – John considers this to be an essential part of the final solution. The development team indicate this feature is similar to the traffic alerting service, which is in scope for the January release, and should, therefore, be fairly easy to develop.

With no proposed changes to timescales or budget, what priority will John's extra requirement receive for the January release?

- A** Must have.
- B** Should have.
- C** Could have.
- D** Want to have, but won't have this time.

- 4** Patricia works for ABC Insurance Services, she has been asked to take on the role of sponsor for the project, which will look at the Claims Assessment System. She is new to the role of sponsor and has been researching her responsibilities.

Which **one** of the following suggested duties required of the project sponsor during requirements engineering is **TRUE**?

- A** Ensure that the project is completed on time.
- B** Assign the approval of the requirements to the project manager.
- C** Allocate work to appropriate people working on the project.
- D** Agree and sign off the project initiation document.

- 5 A Business Analyst working on a project has just been notified of an update to a business policy, which will impact a set of baselined requirements. The Business Analyst will need to consult with the business stakeholders and agree on the change that needs to be made. They will also need to determine if any other related requirements will be impacted.

Which elements of requirements management **WOULD** enable an effective response to this event? Select the **two** that apply.

- A Requirements identification.
- B Change control.
- C Configuration management.
- D Requirements cross-referencing.

- 6 A charity has three business areas: fundraising, information technology and finance. It has commissioned a project to introduce a new online charity donation 'pick-up request' service. The system will allow a person donating items (a user) to request a date for the collection of items they wish to donate. The donated items are then sold to raise funds for the charity.

The following requirements have been identified:

- 1) The user shall be able to enter items they wish to donate and request a collection date.
- 2) The accounts system shall be able to support up to 20 concurrent users.
- 3) The solution shall be compatible with all commonly used browsers.
- 4) The user shall receive a text message confirmation of their collection date and donated items.

Which of the following represents the **CORRECT** categorisation of these requirements by business area?

- A 2 x Functional (Fund Raising).
1 x Non-Functional (Finance).
1 x Technical (Information Technology).
- B 1 x Functional (Finance).
1 x Non-Functional (Fund Raising).
2 x Technical (Information Technology).
- C 1 x Functional (Fund Raising).
1 x Non-Functional (Fund Raising).
1 x General (Information Technology).
- D 1 x Technical (Finance).
2 x Functional (Fund Raising).
2 x Technical (Information Technology).

7 A practice guide for requirements elicitation offers the following information:

Advantages:

- Can provide a basis for developing prototypes.
- Effective way of identifying exception conditions.
- Efficient method of revealing tacit knowledge.

Disadvantage:

- Likely to be complex to develop.

To which elicitation method does this extract refer?

- A Scenario analysis.
- B Storytelling.
- C Workshop.
- D Interview.

8 Kings Castle Travel organises UK walking tours and has begun a programme to upgrade its IT systems. Its Walks System will hold details of all the walking routes the company uses to create its tour packages. These routes are created by the walking guides and are currently held in spreadsheet files on the office computer.

A use case diagram is being developed to model the functionality of the new system. This will include the ability to import walking routes from these spreadsheets. All routes held on the system will be checked by the walking guides and authorised for use by the tour manager.

What will **NOT** be an actor on the use case diagram?

- A Walking guides.
- B Walks system.
- C Spreadsheet walking routes system.
- D Tour Manager.

- 9 A project sponsor is part of a review group, during the requirements validation stage of the requirements engineering framework.

The PRIMARY validation role of the project sponsor is to ensure that the requirements are what at this stage?

- A Technically feasible.
- B Unambiguous.
- C Aligned with business objectives.
- D Compliant with quality standards and policies.

- 10 A Business Analyst is in the process of validating a set of requirements produced for a new business initiative. The following need to be agreed:

- That the delivery of requirements is within budget.
- That the requirements are technically feasible.

Who is responsible for agreeing these?

- A Project manager, developers.
- B Subject matter expert, developers.
- C Project sponsor, testers.
- D Project manager, testers.

- 11 A Business Analyst has just been informed by her project manager that a baselined requirement is likely to be removed or deferred from the set, to complete the project within a shorter timescale.

Which **two** elements of requirements management **SHOULD** be completed in response to this event?

- A Requirements cross-referencing.
- B Change control.
- C Software Support Tools.
- D Requirements identification.

12 The following requirements have been identified by a Business Analyst:

- A. The system must allow users to record the date the order was dispatched from the warehouse.
- B. The system must allow the warehouse manager to view a summary of orders placed by a customer.
- C. The project must be managed in a way that is compliant with Health and Safety regulations.
- D. The interface of the system should be capable of a data transfer rate of 200Mbps.

In the requirements catalogue, which of these **SHOULD** be categorised as functional requirements?

- A** B and C only.
- B** A and B only.
- C** C and D only.
- D** A, B and D only.

13 William is a Business Analyst working on a project and through interactions with stakeholders, he has identified that some further elicitation could be helpful.

He has identified two areas to address:

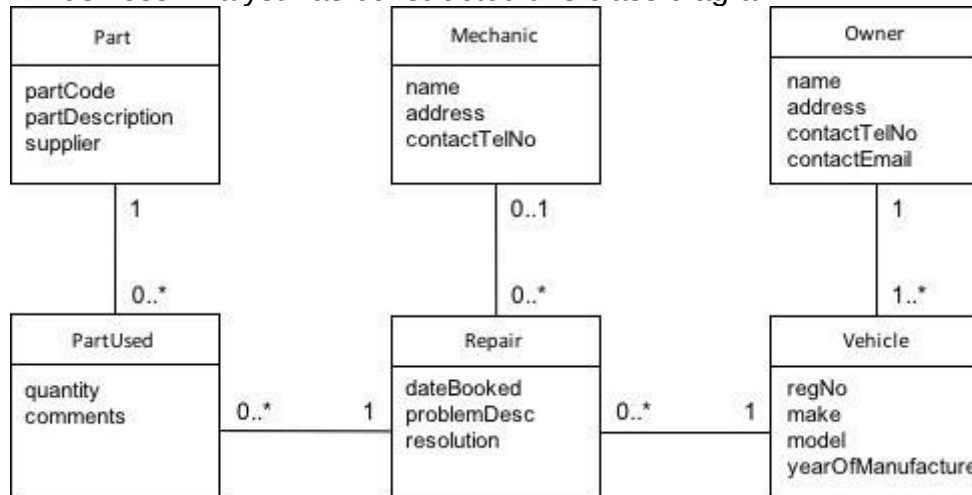
- 1) Stakeholders need to be given more opportunity to discuss requirements.
- 2) Users need to be able to visualise the solution more clearly, to help them communicate their requirements.

Which elicitation technique would **BEST** address both these issues?

- A** Scenario analysis.
- B** Focus group.
- C** Prototyping.
- D** Activity sampling.

- 14 Village Autos is a vehicle repair company specialising in maintaining vintage vehicles. It is in the process of procuring an automated information system to replace the current paper-based processes.

A Business Analyst has constructed this class diagram:



Which of these business rules is **NOT** supported by the class diagram?

- A A mechanic may not be needed to repair the vehicle.
 - B A repair may require many parts to be used.
 - C A vehicle may have more than one owner.
 - D A vehicle may be repaired many times over the years.
- 15 A company is undertaking the requirements engineering for a new product. Some of the requirements are close to the limits of currently available technology. The Business Analyst has approached a senior developer to check the technical feasibility of the requirements.

In which stage of the requirements engineering framework **SHOULD** this action take place?

- A Validation.
- B Analysis.
- C Documentation.
- D Management.

- 16** Hollybush Pet Food Company has initiated a project to introduce more efficient processes into its finance function, supported by a new IT system. It is now time to sign-off the requirements of the project and move into design and Philip, the Business Analyst, would like to hold a formal review of requirements. He has been asked to explain to the project sponsor why this is a good idea.

Which of the following reasons **SHOULD** he give for wanting to hold a formal review?

- A. Allows requirements to be considered from different perspectives.
 - B. Removes the need for any other review of the requirements.
 - C. Allows defects in requirements to be identified before they are moved into design.
 - D. Means that requirements do not have to be documented in detail.
-
- A** A and D only.
 - B** B and C only.
 - C** A and C only.
 - D** C and D only.

- 17** A developer working to implement a set of baselined requirements has called the Business Analyst responsible for the project to check her understanding of a key requirement. During the conversation, it becomes clear to both parties that the developer is working on an older version of the requirement.

Which element of requirements management has failed?

- A** Configuration management.
- B** Change control.
- C** Requirements identification.
- D** Requirements cross-referencing.

- 18** A Business Analyst is documenting the requirements to be developed for an Agile project using user stories. A colleague has suggested that they also consider using data models alongside the user stories.

Which of the following **BEST** describes the reasoning for this recommendation?

- A** Modelling is more accurate than user stories.
- B** User stories can be too challenging for stakeholders to interpret.
- C** Modelling can help to identify any gaps in the user stories.
- D** Modelling would remove the need for further stakeholder input.

- 19** A car manufacturer is working with a Business Analyst to deliver new features on their range of vehicles. Consequently, it has invited customers to meet with the development team to answer a series of questions about the current vehicles and improvements they would like to see.

Which requirements elicitation technique is going to be used?

- A** Storytelling.
- B** Interview.
- C** Document analysis.
- D** Observation.

- 20** A Business Analyst is working on a project which favours an Agile project approach. Which of the following approaches to requirements validation would be considered **MOST** suitable for the analyst to use?

- A** Requirements outlines should be established, with a selected subset of the requirements being elaborated in each iteration.
- B** Requirements outlines should be established and elaborated throughout the first iteration.
- C** Requirements should be fully established at the outset and reviewed at the discretion of the Project Manager.
- D** Requirements outlines should be fully established, and reviewed at the end of the first iteration to define next steps for the project team.

- 21** Mel is a Business Analyst who has returned to a project she had previously been a part of.

She is keen to understand how some of the requirements she elicited early in the project were finally implemented.

Which type of traceability is Mel seeking?

- A** Vertical 'upwards'.
- B** Horizontal 'forwards to'.
- C** Horizontal 'backwards from'.
- D** Vertical 'downwards'.

- 22** A Business Analyst has been working with stakeholders to produce user stories for a new, wearable health monitoring system.

The following user stories have been identified:

- a) As the production manager, I need to produce products which generate revenue and profit for the company.
- b) As a sales representative, I want to view what I have sold because I need to know what my commission will be.
- c) As the regional director, I want to know who my most loyal and lucrative customers are likely to be.
- d) As the chief technology officer, I want to be able to monitor and investigate new developments in health informatics.
- e) As the managing director, I want to know who is selling the most devices and reward them for their efforts.

Which of these do **NOT** include all the required elements of a user story?

- A** b, d and e only.
- B** a, c and d only.
- C** c, d and e only.
- D** a, b and e only.

- 23** Alyssa has been working as a BUSINESS ANALYST on waterfall projects for a number of years and is now embarking on a project that will use an Agile approach.

Which of the following **BEST** describes the stance she will need to take when eliciting requirements?

- A** She can still use many of the techniques that she used on the waterfall projects, but some may require modification for an Agile project.
- B** She will need to use totally different elicitation techniques.
- C** She should only use workshops and prototyping for Agile projects.
- D** In waterfall it wouldn't have been appropriate to prototype, but this is essential in Agile.

- 24** The following extract has been taken from a guide for requirements elicitation:

Advantages:

- Gives direct access to the working environment.
- Provides valid in-depth understanding.
- Exposes tacit knowledge.

Disadvantages:

- May be time-consuming.
- Can be intrusive and therefore affect the situation and hence the requirements elicited.

Which elicitation method is being referred to in the extract?

- A** Workshops.
- B** Interviews.
- C** Focus Groups.
- D** Observation.

- 25** A business manager has queried the need for a particular feature within the organisation's customer account management software. The Business Analyst has been asked to trace the original requirement which has resulted in the development of this feature.

Which area of the Requirements Engineering Framework **SHOULD** this activity fall under?

- A** Analysis.
- B** Validation.
- C** Management.
- D** Elicitation.

- 26** In discussions with a project manager, a Business Analyst has defined these items as “Features or characteristics that have been requested by a stakeholder and may form part of a solution”.

What is the Business Analyst describing?

- A** KPIs.
- B** Actors.
- C** Use Cases.
- D** Requirements.

- 27** Gordon is a Business Analyst who is defining requirements for a new application and wishes to visually demonstrate it to a group of stakeholders to elicit feedback.

Which technique **SHOULD** Gordon use?

- A** Prototyping.
- B** Scenario analysis.
- C** Storytelling.
- D** Interviews.

- 28** A Business Analyst is collaborating with manager of a retail organisation to assess the existing customer returns process. A variety of profiles have been created which detail the individual goals, needs and characteristics of different sample customer types. These profiles will be used to assess the customer journey from a range of perspectives and explore their individual experiences and any areas for improvement.

What is the **CORRECT** term to describe these customer profiles?

- A** User roles.
- B** Personas.
- C** Touchpoints.
- D** Use cases.

- 29** Terry is a Business Analyst working on a project to create a reservation management solution for a hotel chain – this solution will link to the housekeeping and maintenance schedule to ensure maximum availability. Terry has scheduled time for user roles analysis, and has been asked to justify this time to the Project Manager.

Which **two** of the following are benefits of completing user roles analysis?

- A** Identifying common interests and requirements.
- B** Identifying issues with the financial feasibility of the solution.
- C** Creating a strong base for scenario analysis.
- D** Identifying users who are likely to object to the implementation of the solution.

- 30** Ted is a Business Analyst for an architecture firm, who is acting as Project Manager on an internal project, aiming to upgrade the system currently used for submitting and tracking planning applications. Ted is concerned about the ongoing management of the requirements, during the system design and build.

Which of the following is **NOT** a benefit of robust requirements management?

- A** Established traceability.
- B** Greater control of changes.
- C** Reduces the likelihood of future problems.
- D** Ensures accessibility for the end-user.

- 31** Tiffany is a Business Analyst who is focused on ensuring traceability of requirements.

What does this “backwards from” traceability allow Tiffany to do?

- A** To trace the origin of requirements.
- B** To examine the total cost of the requirements.
- C** To ensure the requirement aligns with business objectives.
- D** To remove unnecessary requirements as they become irrelevant.

- 32** Scott has chosen to model the requirements which have been elicited for his current project.

Which of the following is **NOT** a benefit of modelling?

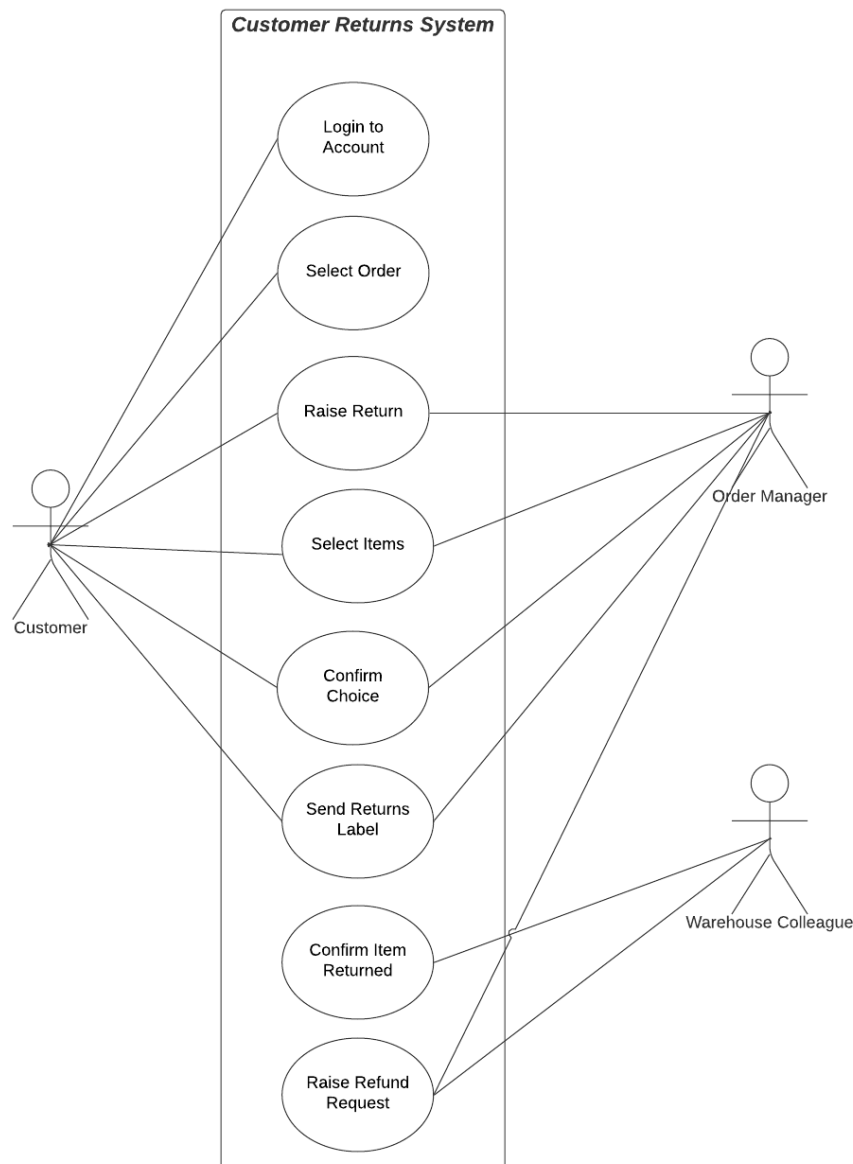
- A** Provides clarity.
- B** Confirms requirements are in scope.
- C** Removes ambiguity.
- D** Confirms priority of requirements.

- 33** Alistair is a Business Analyst who is aiming to understand the business rules associated with the relationships between clients and contracts and the data that can and can't be held. These processes are subject to strict internal and external regulations due to the sensitive nature of the data involved, and must be adhered to.

Which **one** of the following modelling techniques would be **MOST SUITABLE** in helping Alistair define these business rules?

- A** Class diagrams.
- B** Activity diagrams.
- C** Flowchart.
- D** Use case diagrams.

- 34 Sophia is a Business Analyst, who is modelling the current features for an online retailer. There are multiple features to consider and various actors who will interact with the system.



Which of the following business rules is **NOT** implied on the diagram?

- A A warehouse colleague can raise a refund request.
- B A customer can raise a return.
- C An order manager can confirm items are returned.
- D A warehouse colleague can confirm items are returned.

35 A Business Analyst has completed their requirements analysis and is seeking formal validation from the assigned Review Group. Which of the following documents **SHOULD** be presented to the group in this instance?

- A** Project Plans.
- B** Business Requirements Document.
- C** Business Case.
- D** Solution Prototype.

36 Amber is a Business Analyst who has created a sample of different models relating to the development of a solution relating to claims management for an insurance provider. Amber has used a CRUD matrix to cross-check her models.

Which of the following **CORRECTLY** describe Amber's use of a CRUD matrix?

- A** Attempting to understand the system from different viewpoints.
- B** Sharing the key features of the solution.
- C** Establishing testability.
- D** Identifying omissions or errors.

37 Florence is a Business Analyst who has elicited a range of requirements relating to her current project. Florence plans to use a range of analysis techniques to refine these requirements before progressing.

Which of the following would **NOT** be an activity of Florence's analysis?

- A** Evaluate feasibility.
- B** Identify and remove conflicting requirements.
- C** Identify and remove duplication within requirements.
- D** Evaluate effectiveness.

- 38** Janine is a Business Analyst in an Agile environment, who has elicited a number of requirements. The project manager is keen to get development underway, as competitors are developing similar products and therefore the timescale for the first iteration is challenging.

The intention is to come to market as soon as possible, with further development planned on future iterations.

To allow development to begin quickly, what **SHOULD** Janine do with the requirements?

- A** Reprioritise the requirements.
- B** Slice the requirements.
- C** Elaborate as many requirements as possible.
- D** Remove a number of requirements.

- 39** Thomas is a Business Analyst working on a project for the local government, relating to income support eligibility and benefits allowances. As this is a highly regulated sector it is of great importance that any calculations performed within the solution have a very high level of accuracy. Thomas is working to understand the business rules related to these calculations and is considering methods for analysing these rules.

Which type of business rules will need to be in place to ensure the accuracy of the calculations?

- A** External constraints.
- B** Internal policy.
- C** Regulatory fines.
- D** Operational procedures.

- 40** Aaron has been challenged to ensure that upon completion of the project, all requirements must be testable.

Which of the following **BEST** describes the importance of this testability?

- A** To ensure that all requirements have been delivered as intended.
- B** To ensure that all requirements are documented.
- C** To ensure that all requirements are adding value.
- D** To ensure that all requirements align with business objectives.

End of Paper

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Sample Paper – Answer Key

Question	Answer	Explanation / Rationale	Syllabus Sections
1	A	This describes some of the advantages and disadvantages of interviews.	2.3
2	C	A class model would show the minimum and maximum values.	4.2
3	C	This requirement is desirable and may be included in this increment if budget and timescales allow.	6.2
4	D	It is the role of the Project Sponsor to sign off on the project initiation.	5.1
5	B,D	As the requirements have been baselined, formal change control must be requested, and any impact on other requirements must be identified.	8.4
6	A	These requirements span the three business areas, with two being fundraising focused.	3.1
7	A	Scenario analysis can provide the advantages listed, although it can be time consuming to conduct.	2.4
8	B	The system under development would not be an actor in the diagram.	4.3
9	C	The Project Sponsor would be responsible for ensuring alignment between the requirements and the overall business objectives.	5.1
10	A	The project manager will confirm budget alignment, with developers advising on technical feasibility.	5.1
11	A, B	Formal change control must be completed and cross referencing of the requirements should be undertaken due to the status of the requirements.	8.4
12	B	Requirements C and D would be classed as general and technical.	3.1
13	C	Prototyping would allow interaction with the solution and allow for some demonstration of its functions.	2.4
14	C	The diagram shows that although an owner may have unlimited vehicles, each vehicle may only have one owner.	4.4
15	B	This activity would be completed during analysis, to ascertain technical feasibility of requirements.	6.3
16	C	Documentation and time must still be dedicated to the review of requirements.	5.3
17	A	As the requirements have been baselined, this would be classed as configuration management.	8.5
18	C	User stories are often used in agile projects, with models helping to fill any gaps in understanding.	3.2

Question	Answer	Explanation / Rationale	Syllabus Sections
19	B	Interviewing customers will allow them to share their individual views.	2.4
20	A	This approach to validation aligns with the iterative project approach.	5.4
21	B	Horizontal “forwards to” traceability demonstrates how requirements have developed and into what.	8.6
22	B	These requirements do not fulfil each element of the “as a, I want, so that” format.	3.5
23	A	Techniques may be useable but will require adaptation.	2.5
24	D	Observations offers the advantages and disadvantages listed.	2.3
25	C	Traceability is part of requirements “Management”.	1.2
26	D	This is the definition of a requirement according to Business Analysis 4 th Edition.	1.1
27	A	Prototyping would allow for this demonstration and review of the process.	4.6
28	B	Personas are used in this manner, to provide more insight into the customer experience – they allow for more detailed and individual perspective.	7.1
29	A,C	B and D are not linked to user roles analysis.	7.2
30	D	This is not an explicit benefit of requirements management.	8.2
31	A	Backwards from traceability allows the original source of the requirement to be identified.	8.1
32	D	Modelling alone would not prioritise requirements.	4.1
33	A	This modelling technique would be more appropriate in this circumstance.	4.2
34	C	There is a missing association between “order manager” and the “confirm items are returned”.	4.3
35	B	The Business Requirements Document would be used in this instance.	5.5
36	D	Cross checking using a CRUD matrix can help to identify errors.	4.5
37	D	Effectiveness cannot be measured during analysis.	6.1
38	B	Slicing would allow development to commence and requirements to be elaborated in each iteration.	6.4
39	D	These procedures would need to be in place.	6.5
40	A	Testability simply ensures that requirements have been delivered as intended.	6.6